# Probability & Statistics

Books on probability and statistics
https://www.quora.com/What-are-some-good-books-for-learning-probability-and-statistics

Introduction to Probability Theory

Probability Definitions
Probability of an event is the measure of the likelihood that the event will occur (in some experiment).

Probability is a number between 0 and 1.
But sometimes people express probability using percents.

0 – event is impossible
0.5 – the event is just as likely to occur as it is to not occur
1 – event will certainly happen, 100%.

The sum of probabilities of all outcomes of an experiment should always be 1.

Outcomes can be combinations of individual events.

For example, if we throw one dice cube – there are 6 possible outcomes (1,2,3,4,5,6), each one has same probability – 1/6.

If we throw 2 cubes, there are 6*6=36 possible combinations.
Probability of each combination is the same – 1/36.

If we define an outcome as a sum of numbers on top side of cubes after the throw, then there are 11 possible outcomes:

| Sum | Combinations | Number of combinations | Probability |
|---|---|---|---|
| 2 | 1+1 | 1 | 1/36 |
| 3 | 1+2,2+1 | 2 | 2/36 |
| 4 | 1+3,2+2,3+1 | 3 | 3/36 |
| 5 | 1+4,2+3,3+2,4+1 | 4 | 4/36 |
| 6 | 1+5,2+4,3+3,4+2,5+1 | 5 | 5/36 |
| 7 | 1+6,2+5,3+4,4+3,5+2,6+1 | 6 | 6/36 |
| 8 | 2+6,3+5,4+4,5+3,6+2 | 5 | 5/36 |
| 9 | 3+6,4+5,5+4,6+3 | 4 | 4/36 |
| 10 | 4+6,5+5,6+4 | 3 | 3/36 |
| 11 | 5+6,6+5 | 2 | 2/36 |
| 12 | 6+6 | 1 | 1/36 |
| Total | | 36 | 1 |

From the table above you see how some outcomes can result from more combinations, thus have higher probability of happening.

Note: Events can be mutually exclusive, or not.
Also, events may be independent, or dependent on each other.

**Marginal Probability** = Simple Probability , a probability of an independent event occurring.

The term “marginal variable” is used to refer to a subset of variables in a table. They called “marginal” because they used to be calculated by summing values in a table along rows or columns , and writing the sum in the margins of the table.

Examples of assigning probability.

### Example 1. Tossing a coin.

Possible outcomes : (tails or heads).
Events (tails of heads) are mutually exclusive.
Their probabilities should be equal.

$P_t + P_h = 1$
$P_t = 0.5$
$P_h = 0.5$

### Example 2. Rolling a die.

Possible outcomes: (1,2,3,4,5,6).
They are mutually exclusive, probability equal, sum() = 1
So each probability = 1/6 = 0.167 3.

### Example 3. Job Search.

I get 2 emails with job descriptions every day, 10 per week.

Two of them may be a fit: 2 of 10 (0.2).

So I send an email that I am interested.
I am invited to every third position I requested (0.2/3 = 0.067)

And I get 1 position out of 10 initial interviews (0.0067).

So the probability that an incoming email will result in a job is 0.0067, or ~1%.
So, I need to consider 100 positions (this will take 10 weeks), send 20 emails asking to invite, go to 6..7 interviews, and eventually get a job.

So job search takes 2..3 months.

## Rules (Or, And, etc.)

Lets A be an event.
P(A) – probabiltiy of event A
then: P(A) + P(not A) = 1

If A and B are events, then:
P(A or B) = P(A) + P(B) – P(A and B)

for mutually exclusive events P(A and B) = 0,
so P(A or B) = P(A) + P(B)

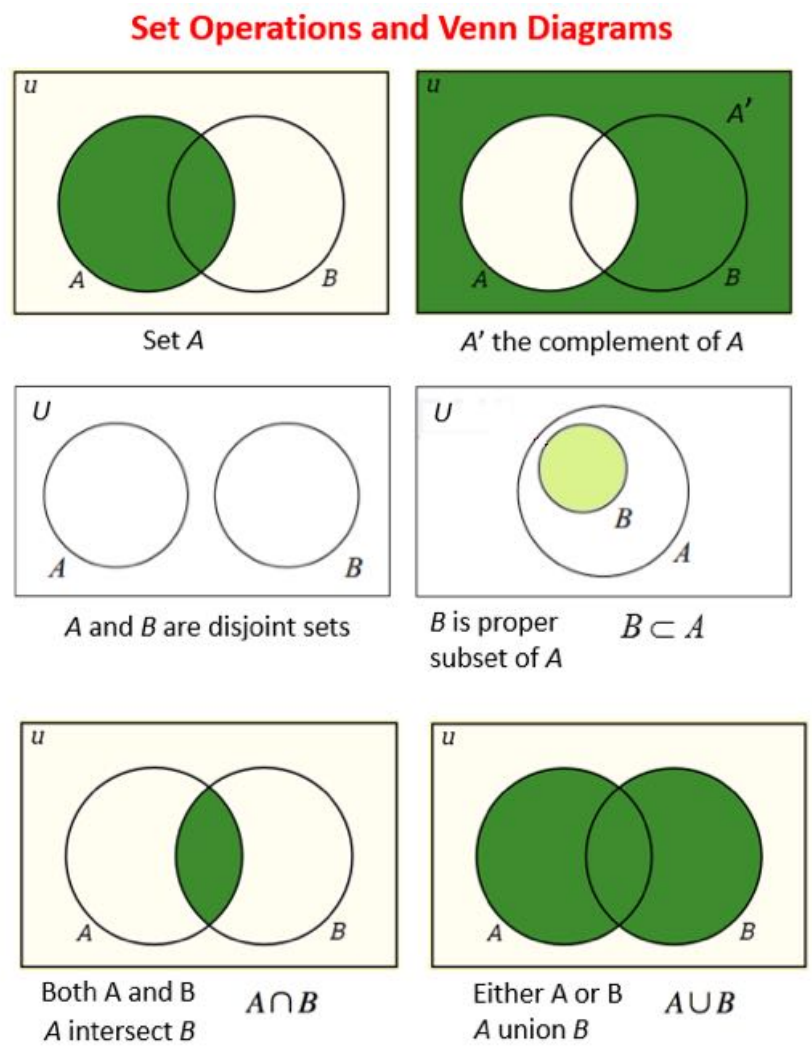
## Venn Diagrams

Venn Diagrams = Graphical representation of events as circles.
See many examples – just google images for: Venn Diagram A or B.

For example, this image shows Intersection (A and B),
negation of intersection (A but not B, or B but not A),
union (A or B), and subtraction (A but not B) :

Conditional Probability
P(A|B) – probability of A given that B has occurred.
P(A|B) = P(A and B) / P(B) (if P(B) is not zero)

If A & B are independent form each other, then:
P(A|B) = P(A)
P(B|A) = P(B)



Bayes' Theorem (Law, Rule)
– https://en.wikipedia.org/wiki/Thomas_Bayes -
– https://en.wikipedia.org/wiki/Bayes%27_theorem –
– https://en.wikipedia.org/wiki/Bayesian_inference –

P(A|B) = P(B|A)*P(A)/P(B)

where A and B are events and P(B) not equal 0.

Frequentist interpretation of Bayes' theorem.
Probability measures a proportion of outcomes.
Suppose an experiment is performed many times.

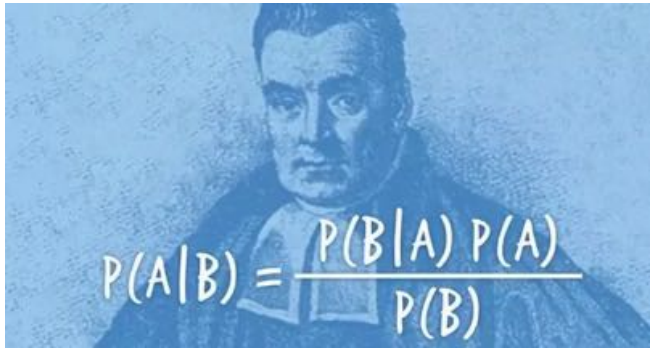
P(A) is the proportion of outcomes with property A,
and P(B) that with property B.

P(B|A) is the proportion of outcomes with property B
out of outcomes with property A,

and P(A|B) the proportion of those with A out of those with B.

Bayesian interpretation of Bayes' theorem.
Probability measures a degree of belief.
Bayes' theorem then links the degree of belief in a proposition before (A)
and after (B) accounting for evidence.

For example, suppose it is believed with 50% certainty
that a coin is twice as likely to land heads than tails.
If the coin is flipped a number of times and the outcomes observed,
then that degree of belief may rise, fall or
remain the same depending on the results.




Thomas_Bayes – (/'be?z/; 1701–1761)

For proposition A and evidence B,
P(A), the prior, is the initial degree of belief in A.
P(A|B), the posterior, is the degree of belief having accounted for B.
P(B|A)/P(B) – represents the support B provides for A.

See also Bayesian inference.
– https://en.wikipedia.org/wiki/Bayesian_inference

Bayesian inference is a method of statistical inference in which
Bayes' theorem is used to update the probability for a hypothesis
as more evidence or information becomes available.

**Bayesian inference -** calculate probability following chains of dependencies or correlations

## Bayes' Theorem (conditional probability)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

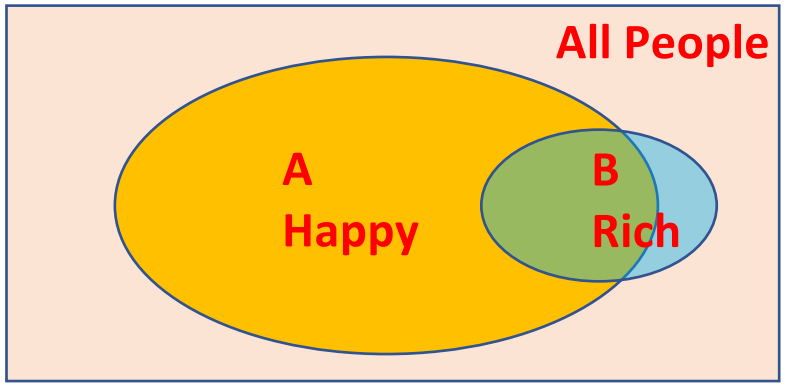
or

$$P(A|B) * P(B) = P(B|A) * P(A)$$

where A and B are events, and P(B) not equal 0.

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

Thomas Bayes
(1701–1761)




Intuitive explanation of Bayesian rule (on the left).
A = Happy, B = Rich
P(A) = 40% = probability of being happy (Yellow)
P(B) = 5% = probability of being rich (Blue)
P(A & B) = 4% = probability of rich & happy (Green)
P(B|A) = 10% = probability of being rich given that Happy
P(A|B) = 80% = probability of being happy given that rich

You can calculate green intersection using two methods:
P(A|B)*P(B) = 80%*5% = 4%
P(B|A)*P(A) = 10%*40% = 4%

## Random Variable. Probability Distribution

**Random variable** is a numerical variable taking different values. Values may be **discrete or continuous**.

For example, think of variable “x” which may have any real value between -10 and +10.

Or think of a variable “z” which can have discrete values 0,0.1,0.2, … 0.9, 1.0

# Discrete Case

**Probability function** P(x) of a discrete random variable “x” is simply a function which gives probability of discrete values.

For example, if the random variable “x” may have two values: ( 0 or 1 ), and if probabilities of those values are equal, then

P(0) = 0.5, f(1) = 0.5.

The sum of individual probabilities for all possible values should be 1.

**Expected value** is an weighted average of all values.

E(x) = sum(Xi*P(Xi)), where Xi goes through all allowed values of X.

**Variance = sigma squared** = sum( (Xi-E(x))**2 * P(Xi) )

**Standard Deviation** = sqrt(sigma squared)

Probability Density Function (PDF)
vs. Cumulative Probability Function ( CPF)

# Continuous Case

**Probability Density Function (PDF)** f(x) is a continuous function.

f(x)*dx gives probability that value is between x and x+dx.

The total integral of f(x)*dx should be equal to 1.

**Cumulative Probability Function (CPF)**
This is simply a function F(x) which gives probability that value is less than x. It monotonously grows from 0 to 1 as x grows (from small to large values).

**The PDF is a first derivative of CPF.**

CPF(x+dx) – CPF(x) = = [Prob. of value < (x+dx)] – [Prob. of value < x] = f(x)*dx

**Mean value = Expected value**

For discrete case we had:
E(x) = sum(Xi*P(Xi))

For continuous case this translates into
E(x) = Integral(xF(x)dx)

## Binomial Distribution

– https://en.wikipedia.org/wiki/Binomial_distribution

if:
Experiment consists of N trials.
Each trial can have two outcomes (True of False).
The probabilities of these outcomes are the same for all trials (p and q = 1-p).
The trials are independent from each other.

Then:
The probability of having "x" True values (out of N) is:

f(x) = C(N,x)*p^x*(1-p)^(N-x)

where C(N,x) = N!/(x!*(N-x)!) – a binomial coefficient

At large numbers the distribution f(x) has a bell-curve shape

**Mean value = Expected value** E(x) = N*p
**Variance** = N*p*(1-p).

## Combinatorics

### Factorial

N! = N*(N-1)*(N-2)*…*1

## Reordering

Let's say, we have N objects (N-element unordered set).

Question: In how many ways can we arrange them?

Answer: N!

Explanation:
N variants for the 1st element (N-1) variants for the 2nd element etc.
Example: How many ways are there to arrange the letters ABCDE in a row?

Answer: 5!=120

## Subsets

Question: In how many ways can we select m-elements out of N-element set?

Answer: C(N,m) = N!/(m!*(N-m)!) – a binomial coefficient

Explanation:
N variants for the first element of subset
(N-1) for the 2nd etc.
(N-m+1) for the m-th element
The above can be written as N!/(N-m)!
Then we need to divide by the number of reordering permutations within the subset, that is by m!

Poisson and Exponential Distributions
– https://en.wikipedia.org/wiki/Poisson_distribution
– https://en.wikipedia.org/wiki/Exponential_distribution

Suppose you are sitting near the road.
You notice that on average you see 10 cars passing by every minute
(we say: the rate lambda = 10/min).
Knowing the average rate, what would be the probability
of having only 5 cars in one minute?
Or 15 cars? Or any given number?

**Poisson Probability** P(k) is the probability of a given
number of events occurring in a fixed interval (of time)
if these events occur with a known average rate
and independently of each other
and of the time since the previous events.

Poisson Formula

P(x) = (lambda**x)*exp(-lambda)/x!

For example, for x=0 (probability of zero events in given interval):

P(0) = (lambda**0)*exp(-lambda)/0! = 1*exp(-lambda)/1 = exp(-lambda)

## Poisson Distribution

Intervals between Poisson events distributed exponentially.

If we have events happening which follow Poisson Distribution,
we can measure the time intervals between these events.

Question: How these times will be distributed?

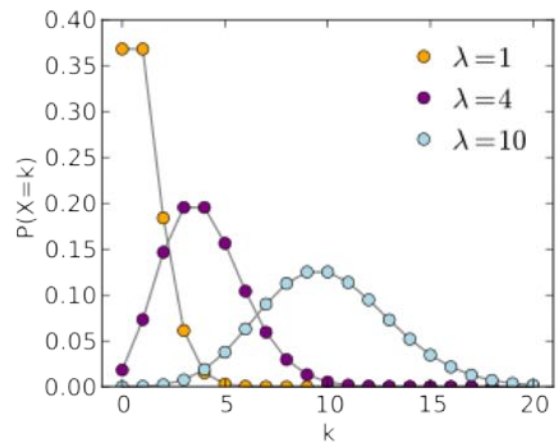
Answer: These times will have exponential distribution
with Probability Density Function (PDF) as:

f(t) = lambda*exp(-lambda*t)

The average time interval will be (1/lambda).
Standard deviation happens to be the same (1/lambda).

$$\lambda = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

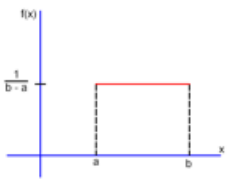


# Uniform and Normal Distributions.

## Uniform Distribution

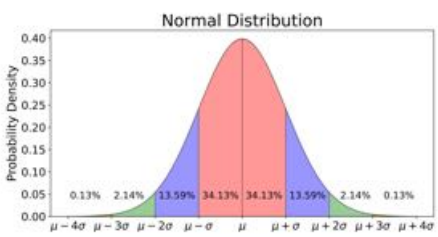
If some value can change between some minimum and maximum values a & b, and the probability of any value in interval [a,b] is the same, then we say that we have uniform distribution.

The PDF (Probability Density Function) for Uniform Distribution looks like a rectangle with the height (1/(b-a)), so that the total area under the curve is exactly 1.




## Normal (Gaussian Bell-shaped) Distribution

The basic shape of the PDF is exp(-x*x/2)
You need to add a coefficient to normalize it to 1
(means – make area under the curve equal to exactly 1).
You can also have the curves have different position
(average value) and different width (sigma).




## Central Limit Theorem (CLT).

The normal distribution is very common in nature.
Because when we have big number of sets of variables,
the distribution of parameters of these sets (for example, averages)
tend to converge to this bell form.
This fact is called **Central Limit Theorem (CLT)**.

Here is a simple example of CLT:

Suppose we have N1 sets, each set has N2 values, all values drawn from the same distribution. The shape of distribution may be anything (uniform, bell-curve, triangular, exponential, etc.).

We calculate an average in each set – so we have N1 average values.
Question: How will these average values be distributed?

Answer: As N1 & N2 become big enough (30+), the distribution will get close to the Normal Gaussian bell curve.

There are some nice animated demonstrations of CLT - just google for them.

## De Moivre–Laplace theorem

De Moire-Laplace theorem is a simple demonstration how a symmetrical binomial distribution can be approximated by Gaussian bell curve (under certain conditions).
See video demonstration below.

https://en.wikipedia.org/wiki/De_Moivre–Laplace_theorem

Introduction into Statistics

"Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data."

Being a Statistician is the major part of being a Data Scientist.
And it is tricky, because there is easy to misuse statistics:
**"There are three kinds of lies: lies, damned lies, and statistics."**

Statistics involves

- Collecting (sampling) the data from different kind of observations
- Visualizing the data, estimating the parameters (average/mean values, errors, etc.)
- Testing validity of a hypothesis (null hypothesis) – or alternative hypothesis
- Estimating intervals, estimating significance of observed differences, etc.

**Here are some basic formulas:**

**formula to calculate mean**

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

**formula to estimate sample standard deviation**

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

**formula to calculate sample SD variance**

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Intuitive explanation for dividing by (n-1)
when calculating standard deviation:

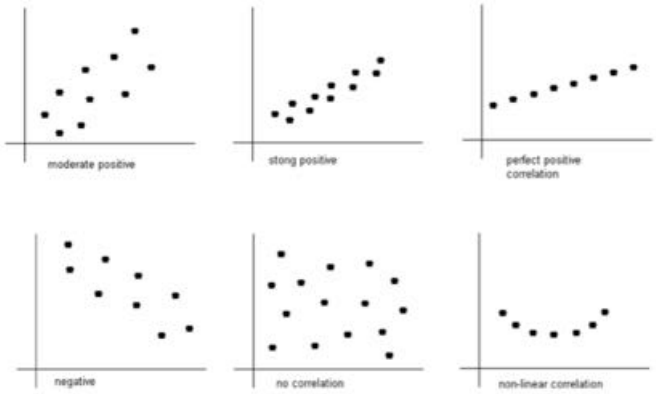
– http://stats.stackexchange.com/questions/3931/intuitive-explanation-for-dividing-by-n-1-when-calculating-standard-deviation

# Correlation and Regression

Pearson Correlation **r**

https://en.wikipedia.org/wiki/Pearson_correlation_coefficient

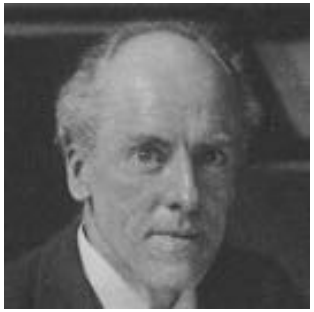
$$r = \frac{SP}{\sqrt{SS_X SS_Y}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$$




$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

Pearson correlation is also known under names:

- Pearson correlation coefficient (PCC)
- Pearson's r
- Pearson product-moment correlation coefficient (PPMCC)
- Bivariate correlation



**Karl Pearson**
(1857 - 1936)
Famous English mathematician and biostatistician

**r** can vary from -1 to 1:
-1 – perfect negative correlation
+1 – perfect positive correlation

$R^2$ is the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)

# Linear Regression

Linear Regression = draw a line through experimental points.

- https://en.wikipedia.org/wiki/Linear_regression
- https://en.wikipedia.org/wiki/Simple_linear_regression

We derive linear model by minimization the mean square error (MSE):

$$\min_{\hat{\beta}_0,\hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

We take partial derivatives by two coefficients - and solve system of 2 equations, and after some algebraic transformations, we derive the standard **OLS** method (**OLS = Ordinary Least Squares**).

$y_m$ = y_mean = sum(y)/N
$x_m$ = x_mean = sum(x)/N

var(x) = variance(x) = sum($(x-x_m)^2$)
covar(x,y) = covariance(x,y) = sum($(y-y_m)*(x-x_m)$)

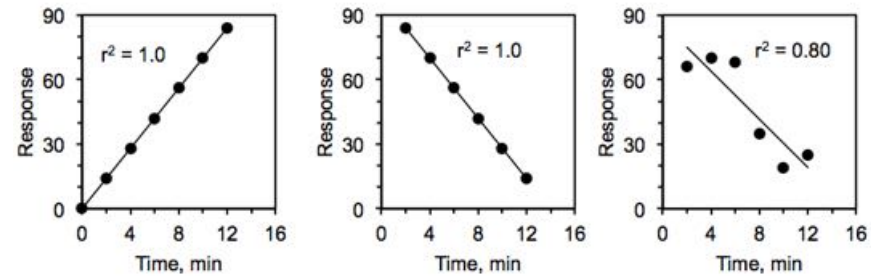
slope $b_1$ = covar(x,y) / var(x)
intercept $b_0$ = $y_m$ - $b*x_m$

When drawing a curve through experimental points,
it is very important to estimate how good this curve
fits the data, and if fitting the data with this curve makes sense.
For this you can use the **Coefficient of Determination $R^2$**:

– https://en.wikipedia.org/wiki/Coefficient_of_determination

"R squared" is a number between 0 and 1.
$R^2$ close to 1 – good model.

$R^2$ is the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)




**Coefficient of Determination (R Square)**

$$R^2 = \frac{SSR}{SST}$$

$$SSR = \sum_i (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_i (y_i - \bar{y})^2$$

Where,

- SSR is Sum of Squared Regression also known as variation explained by the model
- SST is Total variation in the data also known as sum of squared total
- y_i is the y value for observation i
- y_bar is the mean of y value
- y_bar_hat is predicted value of y for observation i

# Central Limit Theorem (CLT)

We have all heard about Normal (Gaussian) distribution.

This distribution is very foundational in statistics. And it comes from CLT.

Classical CLT mathematically proves that if we calculate group averages in many groups, then the distribution of those averages converges to Normal distribution as long as we have large number of points in each group (30..100 or more) and large number of groups (30-100 or more).

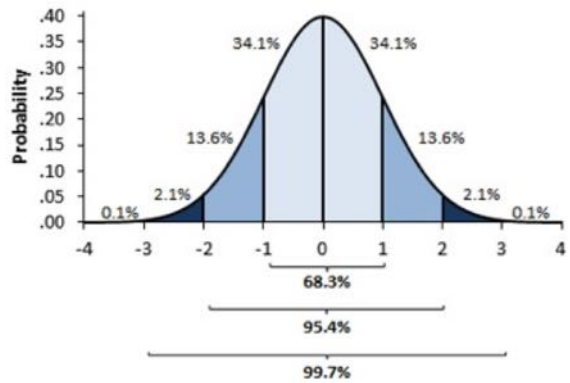
Specifically, suppose we have our data points sampled from a distribution with finite average and finite variance. We subdivide this data into Ng groups, each containing Ns samples:

[...] [...] [...] [...] [...] [...] [...] [...] [...]

For each group we calculate average value of its Ns samples. Thus we have Ng averages values, and we plot them as a histogram.

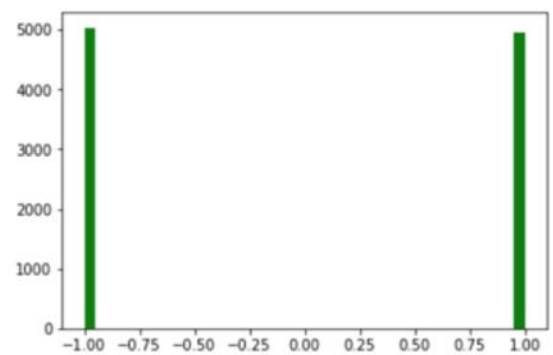
If Ng and Ns are big enough, the histogram will converge to a normal distribution. This holds true regardless of the shape of the original distribution of data.

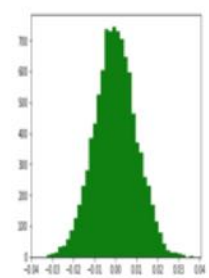
For example, suppose we have a million data points, each value is either 1 or -1 (random coin toss). We split data into Ng=10,000 groups, each sample group has Ns=10,000 data points. Then in each group we calculate the mean value (which will be close to zero).
Now we have 10,000 of mean values - and we make a histogram of those values. According to CLT it will be bell-shaped.




$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\mu =$ Mean
$\sigma =$ Standard Deviation
$\pi \approx 3.14159\cdots$
$e \approx 2.71828\cdots$



# p-value & null hypothesis

A **p-value** is a probability of getting results at least as extreme as yours. If this probability is very small, this means that the result is unlikely to be explained by the null hypothesis. So we should reject the null hypothesis.

```
P-value    probability of observed
-------------------------------
P < 0.05         5%
P < 0.01         1%
p < 0.001        0.1%
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```
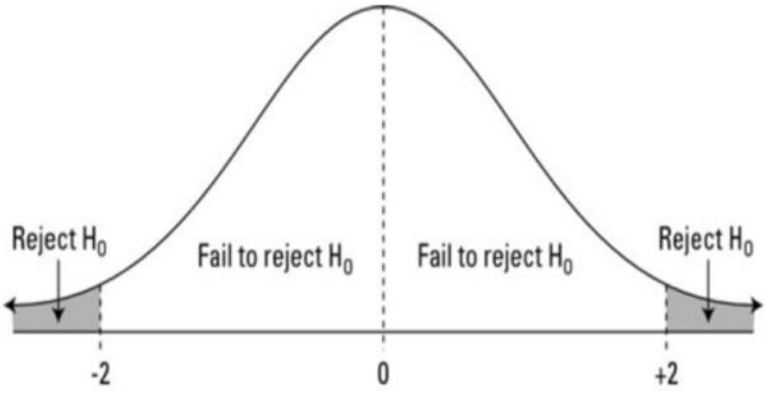

A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you reject the null hypothesis. People say that this evidence is "statistically significant".

Related term is **Confidence Interval (CI)**. A 95% CI is the interval that you are 95% certain contains the true population value as it might be estimated from a much larger study.

**z-score** is the distance from the mean measured in the number of standard deviations.




Type I error - false rejection of the null hypothesis
Type II error - false acceptance of the null hypothesis

| \ Decision<br>Truth \ | Decision to Accept Ho | Decision to Reject Ho |
|---|---|---|
| Ho is True | Correct Decision | type I error |
| Ho is False | type II error | Correct Decision |

Problem1:

Given: women's heights are normally distributed with the mean 65” and standard deviation of 3.5”
Question: What is the probability of finding a random sample of 50 women with a mean height of 70”.

To answer, let's estimate the **z-score**:
$z = (x – \mu) / (\sigma / \sqrt{n}) = (70 – 65) / (3.5/\sqrt{50}) = 5 / 0.495 = 10.1$

We know that 99% of values fall within $3\sigma$ (68, 95, 99.7 rule).
We have z-score = 10, which is much more than 3.
So the probability that any sample of 50 women will have a mean height of 70” will be much-much less than 1%.

Problem 2:

Suppose you are testing a claim that the percentage of all women with varicose veins is 25%, and your sample of 100 women had 20% with varicose veins.
Then the sample proportion p=0.20.
The standard error for your sample percentage is the square root of p(1-p)/n which equals 0.04 or 4%.

You find the test statistic by taking the proportion in the sample with varicose veins, 0.20, subtracting the claimed proportion of all women with varicose veins, 0.25, and then dividing the result by the standard error, 0.04.

These calculations give you a test statistic (standard score) of –0.05 divided by 0.04 = –1.25.

This tells you that your sample results and the population claim in H0 are 1.25 standard errors apart; in particular, your sample results are 1.25 standard errors below the claim.

When testing H0: $p = 0.25$ versus Ha: $p < 0.25$, you find that the p-value of -1.25 by finding the probability that Z is less than -1.25. When you look this number up on the **Z-table**, you find a probability of 0.1056 of Z being less than this value.

Note: A **Z-table** is simply a table of values of the cumulative distribution function of the normal distribution.

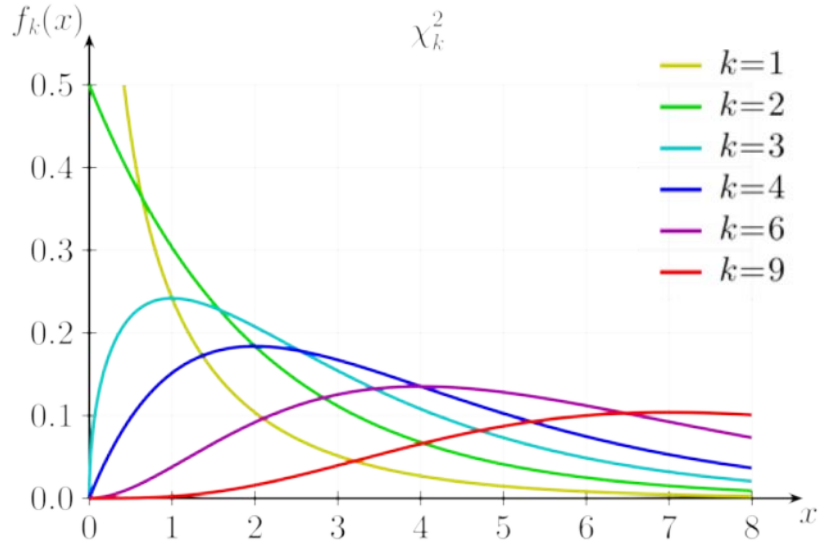
# Chi-Square $\chi^2$ Distribution

If $Z_1, \ldots, Z_k$ are independent, standard normal (Gaussian) random variables, then the sum of their squares:

$$\text{sum}_k(Z_i^2)$$

is distributed as $\chi^2$, with "k" degrees of freedom

Mean = K
variance = 2K

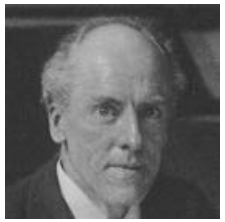



$$f(x;\, k) = \begin{cases} \dfrac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

## Why is it called $\chi 2$ ?

Greek character Chi ( $\chi$ ) appears to be the notation used by Karl Pearson in 1900:

"On the Criterion that a Given System of Deviations from the Probable in the Case of a Correlated System of Variables is such that it can be Reasonably Supposed to have Arisen from Random Sampling," Philosophical Magazine, 50, (1900), 157-175.



Karl Pearson
1857 – 1936
English mathematician
and biostatistician

**Where Gamma Function:**

$$\Gamma(n) = (n-1)!$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad \Re(z) > 0$$

# Chi-Square $\chi^2$ Goodness of Fit Test

https://stattrek.com/chi-square-test/goodness-of-fit.aspx

Pearson, 1900.
Suppose we have k mutually-exclusive classes
N - total number of all observations,
$n_i$ (i=1,2,...k) - number of observations for each class

$p_i$ - hypothetical probability of each class (null hypothesis).
So sum($p_i$) = 1

Then for each class:
expected value of observations $m_i = N*p_i$
sum($m_i$) = N*sum($p_i$) = sum($n_i$)

Karl Pearson has proposed that as N gets bigger, the value

$X^2$ = sum($(n_i-m_i)^2/m_i$) --> $\chi^2$ distribution with (k – 1) degrees of freedom

## Notation

χ2 is the lower-case Greek letter Chi
$f_o$ is the Observed Frequency
$f_e$ is the Expected Frequency

## Chi-Square Statistic

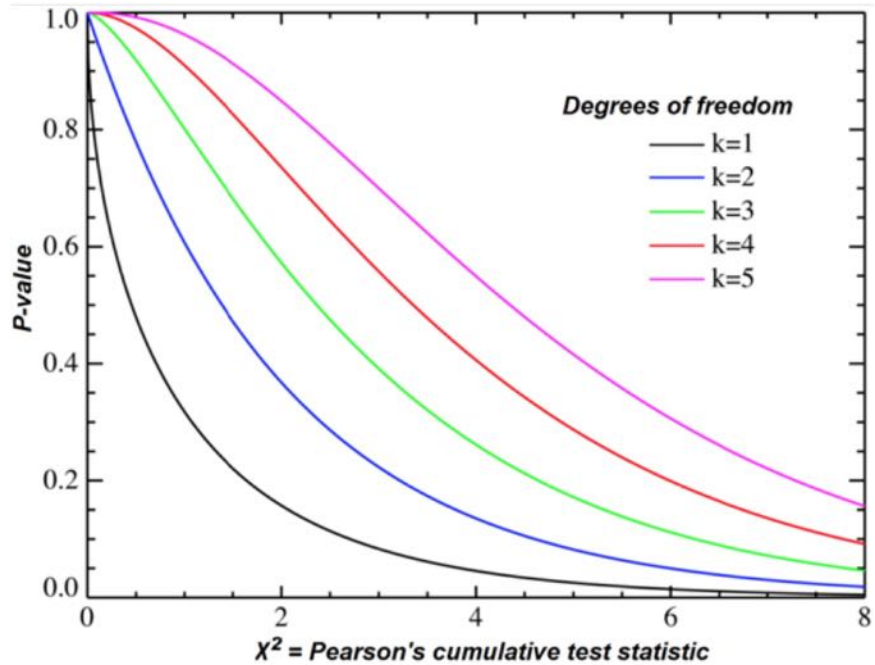
$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

## Computing Expected Frequencies

fc - frequency total for the column
fr - frequency total for the row

$$f_e = \frac{f_c f_r}{n}$$

Chi-squared test, showing χ2 on the x-axis and p-value on the y-axis.




In practice we calculate value $X^2$ from our data, then use $\chi^2$ distribution to estimate the P-value (the probability that a chi-square statistic is more extreme than $X^2$ value that we have.

# Example of $\chi^2$ test ( from https://stattrek.com/chi-square-test/goodness-of-fit.aspx )

**Problem:**

We have 3 types of cards (T1,T2,T3) with theoretical frequency of (30%, 60%, 10%).
We have a random sample of 100 cards with counts (50, 45, 5).
Is this sample consistent with theory? Use a 0.05 level of significance.

**Solution:**

Null hypothesis: The proportions are (30%, 60%, 10%)
Alternative hypothesis: At least one of the proportions in the null hypothesis is false.
Degrees of Freedom DF = k-1 = 3-1 = 2
Expected values:
$E_1 = 100*0.3 = 30$, $E_2 = 100*0.6 = 60$, $E_3 = 100 * 0.10 = 10$
$X^2 = \Sigma [ (O_i - E_i)^2 / E_i ] = [ (50 - 30)^2 / 30 ] + [ (45 - 60)^2 / 60 ] + [ (5 - 10)^2 / 10 ] = 19.58$

The P-value is the probability that a chi-square statistic having 2 degrees of freedom is more extreme than 19.58.
Using Chi-Square Distribution we find P(X2 > 19.58) = 0.0001.

Since the P-value (0.0001) is less than the significance level (0.05), we reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate.
Specifically, the approach is appropriate because the sampling method was simple random sampling, the variable under study was categorical, and each level of the categorical variable had an expected frequency count of at least 5.

# Chi-Square $\chi^2$ Test for Independence

## (two-way tables, association)

**Evaluates the relationship between two variables.**
**It is a non-parametric test, performed on categorical data.**

**Example:**
**500 boys and girls are asked which is their favorite color: blue, green, or pink (see results).**
**Using alpha = 0.05, would you conclude that there is a relationship between gender and favorite color?**

**Degrees of Freedom df = (rows-1)(columns-1) = (2-1)(3-1) = 2**

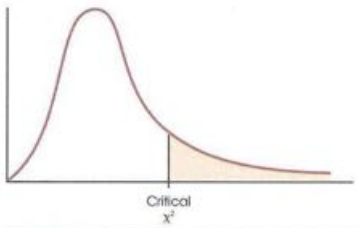
**In $\chi^2$ table the value for alpha=0.05 and df=2 is 5.99**

**Calculate statistics: $\chi^2$ = sum $(f_o - f_e)^2 / f_e$ = 276.39 >> 5.99**

**Conclusion: yes, there is a relationship**

| OBSERVED | Blue | Green | Pink | |
|---|---|---|---|---|
| Boys | 100 | 150 | 20 | 300 |
| Girls | 20 | 30 | 180 | 200 |
| | 120 | 180 | 200 | N = 500 |

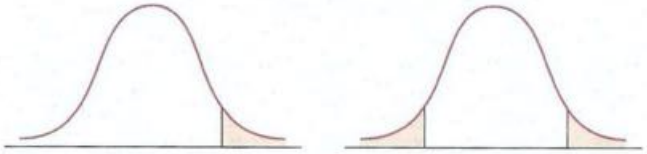
| EXPECTED | Blue | Green | Pink | |
|---|---|---|---|---|
| Boys | 72 | 108 | 120 | 300 |
| Girls | 48 | 72 | 80 | 200 |
| | 120 | 180 | 200 | N = 500 |




## $\chi^2$

| df | Proportion in Critical Region 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|
| 1 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | 4.61 | 5.99 | 7.38 | 9.21 | 10.60 |
| 3 | 6.25 | 7.81 | 9.35 | 11.34 | 12.84 |
| 4 | 7.78 | 9.49 | 11.14 | 13.28 | 14.86 |
| 5 | 9.24 | 11.07 | 12.83 | 15.09 | 16.75 |
| 6 | 10.64 | 12.59 | 14.45 | 16.81 | 18.55 |
| 7 | 12.02 | 14.07 | 16.01 | 18.48 | 20.28 |
| 8 | 13.36 | 15.51 | 17.53 | 20.09 | 21.96 |
| 9 | 14.68 | 16.92 | 19.02 | 21.67 | 23.59 |
| 10 | 15.99 | 18.31 | 20.48 | 23.21 | 25.19 |
| 11 | 17.28 | 19.68 | 21.92 | 24.72 | 26.76 |
| 12 | 18.55 | 21.03 | 23.34 | 26.22 | 28.30 |
| 13 | 19.81 | 22.36 | 24.74 | 27.69 | 29.82 |
| 14 | 21.06 | 23.68 | 26.12 | 29.14 | 31.32 |
| 15 | 22.31 | 25.00 | 27.49 | 30.58 | 32.80 |
| 16 | 23.54 | 26.30 | 28.85 | 32.00 | 34.27 |
| 17 | 24.77 | 27.59 | 30.19 | 33.41 | 35.72 |
| 18 | 25.99 | 28.87 | 31.53 | 34.81 | 37.16 |
| 19 | 27.20 | 30.14 | 32.85 | 36.19 | 38.58 |
| 20 | 28.41 | 31.41 | 34.17 | 37.57 | 40.00 |
| 21 | 29.62 | 32.67 | 35.48 | 38.93 | 41.40 |
| 22 | 30.81 | 33.92 | 36.78 | 40.29 | 42.80 |
| 23 | 32.01 | 35.17 | 38.08 | 41.64 | 44.18 |
| 24 | 33.20 | 36.42 | 39.36 | 42.98 | 45.56 |
| 25 | 34.38 | 37.65 | 40.65 | 44.31 | 46.93 |
| 26 | 35.56 | 38.89 | 41.92 | 45.64 | 48.29 |
| 27 | 36.74 | 40.11 | 43.19 | 46.96 | 49.64 |
| 28 | 37.92 | 41.34 | 44.46 | 48.28 | 50.99 |
| 29 | 39.09 | 42.56 | 45.72 | 49.59 | 52.34 |
| 30 | 40.26 | 43.77 | 46.98 | 50.89 | 53.67 |
| 40 | 51.81 | 55.76 | 59.34 | 63.69 | 66.77 |
| 50 | 63.17 | 67.50 | 71.42 | 76.15 | 79.49 |
| 60 | 74.40 | 79.08 | 83.30 | 88.38 | 91.95 |
| 70 | 85.53 | 90.53 | 95.02 | 100.42 | 104.22 |
| 80 | 96.58 | 101.88 | 106.63 | 112.33 | 116.32 |
| 90 | 107.56 | 113.14 | 118.14 | 124.12 | 128.30 |
| 100 | 118.50 | 124.34 | 129.56 | 135.81 | 140.17 |

## $t$




| df | Proportion in One Tail 0.25 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|---|
| | Proportion in Two Tails Combined 0.50 | 0.20 | 0.10 | 0.05 | 0.02 | 0.01 |
| 1 | 1.000 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 |
| 2 | 0.816 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 |
| 3 | 0.765 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 |
| 4 | 0.741 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 |
| 5 | 0.727 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 |
| 6 | 0.718 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 |
| 7 | 0.711 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 |
| 8 | 0.706 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 |
| 9 | 0.703 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 |
| 10 | 0.700 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 |
| 11 | 0.697 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 |
| 12 | 0.695 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 |
| 13 | 0.694 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 |
| 14 | 0.692 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 |
| 15 | 0.691 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 |
| 16 | 0.690 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 |
| 17 | 0.689 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 |
| 18 | 0.688 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 |
| 19 | 0.688 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 |
| 20 | 0.687 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 |
| 21 | 0.686 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 |
| 22 | 0.686 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 |
| 23 | 0.685 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 |
| 24 | 0.685 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 |
| 25 | 0.684 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 |
| 26 | 0.684 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 |
| 27 | 0.684 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 |
| 28 | 0.683 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 |
| 29 | 0.683 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 |
| 30 | 0.683 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 |
| 40 | 0.681 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 |
| 60 | 0.679 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 |
| 120 | 0.677 | 1.289 | 1.658 | 1.980 | 2.358 | 2.617 |
| ∞ | 0.674 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 |

# Student's *t*-distribution

https://en.wikipedia.org/wiki/Student%27s_t-distribution

Student's t-distribution (or simply the t-distribution)
is any member of a family of continuous probability distributions
that arises when **estimating the mean of a normally distributed population**
in situations **where the sample size is small**
and the population standard deviation is unknown.
It was developed by **William Sealy Gosset** (1908)
under the pseudonym **Student**.

Let Xi be independently and identically drawn from the normal distribution with expected mean value **μ** and variance **σ²**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

----- Sample mean

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

----- Sample variance

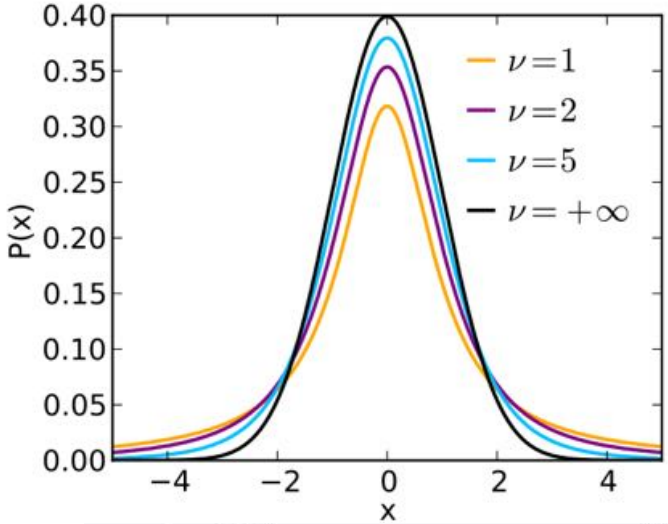
$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

----- Random variable, Normal distribution

$$\frac{\bar{X}-\mu}{S/\sqrt{n}};$$

----- Random variable, **t-distribution**. It uses sample variance (because real sigma is unknown). This creates **uncertainty** – and leads to the "**fat tails**" in the t-distribution.

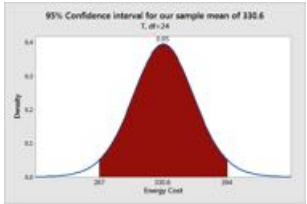
**William Sealy Gosset** used pseudonym "Student" because he was working for **Guinness Beer Company** in England as a "**Head Experimental Brewer**" (working on cultivating barley), and the company didn't allow him to publish under his own name.




$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

In Excel: T.DIST(x,DF,FALSE)
where DF = Degrees of Freedom (Npoints-1)

95% confidence interval



# t-statistics vs z-statistics

z-statistics:

When sample size is big ( > 30 ), then in many cases sample distribution is close to gaussian. So we use gaussian distribution and z-statistics.

Suppose that variable x follows normal gaussian distribution with mean value **μ** and standard deviation **σ**. Then we can compose variable "z" which would follow standard normalized distribution with zero mean and standard deviation = 1 like this:

$$z = (x-\mu)/\sigma$$

Z-score is simply the z-value which shows how far is the value from the mean expressed in "sigmas".
For example, if z > 2, that means that the original value is more than two sigmas away from the mean value.
For normal distribution this means that the probabilty of this even is less than 0.05.
(to be precise, the exact z-score for 0.05 is 1.96)

t-statistics:

When sample size is small ( N < 30 ), then even if we know that the underlying distribution is gaussian, we still can not evaluate the distribution's sigma with enough precision. In this case we need to use t-distribution and t-statistics.

$$t = (x-\mu)/s$$

Here **s** is a sample mean.

t-distribution has "fat tails", so to get 95% confidence interval we need to use t-score larger than 2. How much larger depends on how many data points we have in the sample. **df = degrees of freedom = N-1**

```
N-1    95% t-score
------------------
 1   : 12.71
 2   :  4.30
 3   :  3.18
 5   :  2.57
10   :  2.23
20   :  2.09
30   :  2.04
60   :  2.00
------------------
```

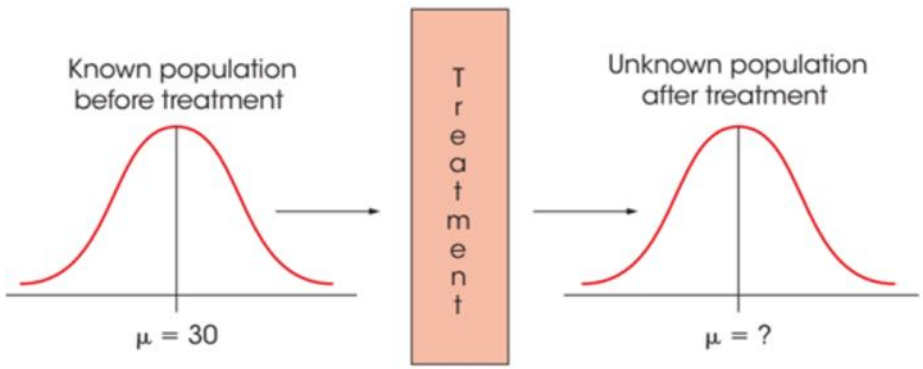
```
from scipy import stats
# Student, p<0.05, 2-tail, same as Excel TINV(0.05,n-1)
for df in [1,2,3,5,10,20,30,60]:
    t_score = stats.t.ppf(1-0.025, df)
    print(f"{df:2}   : {t_score:5.2f}")
```

# Hypothesis Tests with the *t* Statistic

The one-sample t test statistic (assuming the null hypothesis is true)

$$t = \frac{\text{sample mean} - \text{population mean}}{\text{estimated standard error}} = \frac{M - \mu}{s_M} = 0$$

Basic Research Situation for t Statistic:




Assumed that μ is known for the population before treatment.
The purpose of the research study is to determine whether the treatment has an effect.
Note that the population after treatment has unknown values for the mean and the variance. So we will use a sample to test a hypothesis about the population mean.

Four (4) Steps of Hypothesis Testing:

1. State the null hypotheses (H0) and alternative hypotheses, and select an alpha level
2. Locate the critical region using the t-distribution table and df
3. Calculate the t test statistic
4. Make a decision regarding H0 (null hypothesis)

For example:

1. N0: no effect (same mean), two-tailed test, alpha = 0.01
2. For sample size 100, critical t-values = +/-2.6264
3. t-statistics = (sample_mean – population_mean) / s , where s is an estimated standard error (calculated from sample)
4. if || t-statistics || > critical value, then we reject the H0

# t-test – continued

## Effect size.

In addition to rejecting/accepting a Hypothesis, you can also estimate the size of the effect:

- **Cohen's d** - for comparing two samples/groups with same sizes and similar SDs.
- **Glass's delta** - uses only the SD of the control group.
- **Hedges' g** - for two groups with different sample sizes.

Example: for two groups with mean M and standard deviations SD :

**Cohen's d** = $(M_2 - M_1)/SD_{pooled}$

where

$SD_{pooled} = \sqrt{((SD_1^2 + SD_2^2)/2)}$

Determining the amount of variability in scores explained by the treatment effect is an alternative method for measuring effect size.

$$r^2 = \frac{\text{variability accounted for}}{\text{total variability}} = \frac{t^2}{t^2 + df}$$

r2 = 0.01 small effect
r2 = 0.09 medium effect
r2 = 0.25 large effect

## Estimating m from sample

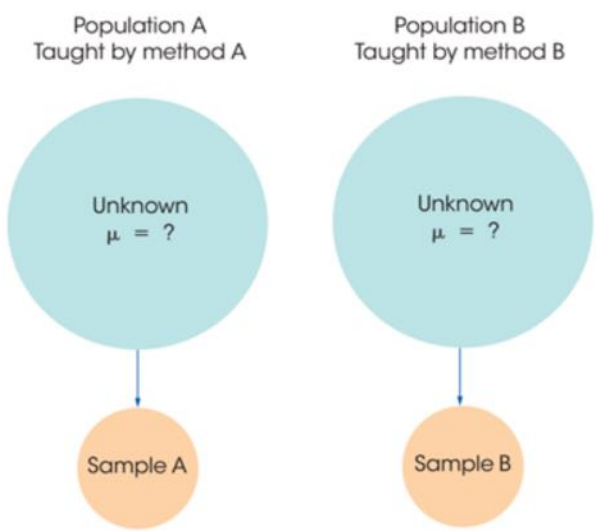
(use sample mean and sample deviation)

$$t = \frac{M - \mu}{s_M} \quad \Rightarrow \quad \mu = M \pm t s_M$$

For any α we know that (1 – α) proportion of t values fall between ± t for the appropriate df.

# The t Test for Two Independent Samples

Data comes from two completely different, independent groups.
Observations within each sample must be independent.
The two populations must be normal.
The two populations must have equal variances (called homogeneity of variance).




Null hypothesis for independent-measures test:

$$H_0 : \mu_1 - \mu_2 = 0$$

Basic structure of the t statistic

$$t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{s_{(M_1 - M_2)}}$$

where estimated standard error (unbiased if n1=n2)

$$s_{(M_1 - M_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

unbiased estimate:

$$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

SS = Sum of Squares
Degrees of freedom df = $df_1 + df_2 = (n_1 - 1) + (n_2 - 1)$

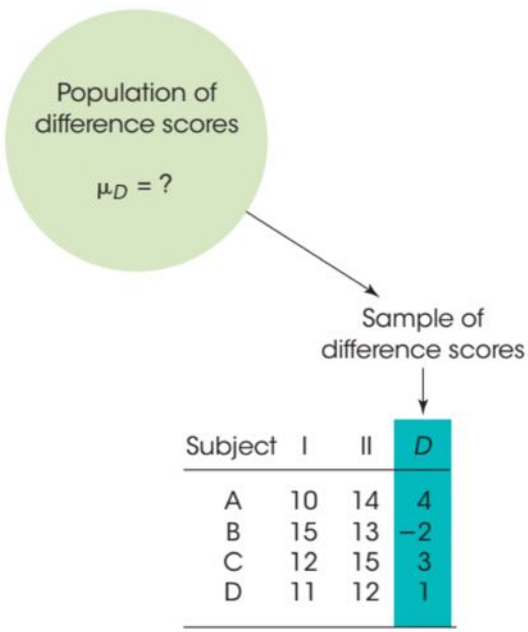
Test for homogeneity of variance:
Levene's test or Brown–Forsythe test
https://en.wikipedia.org/wiki/Levene%27s_test

# The t Test for Repeated-Measures

Repeated-measures design
(a.k.a. within-subjects design)

Two separate scores are obtained **for each individual in the sample**

Same subjects are used in both groups.
Observations within each treatment condition must be independent.
Population distribution of difference scores (D values) must be normally distributed.




| Subject | I | II | D |
|---|---|---|---|
| A | 10 | 14 | 4 |
| B | 15 | 13 | −2 |
| C | 12 | 15 | 3 |
| D | 11 | 12 | 1 |

Difference score = $D = X_2 - X_1$

Mean Difference: $M_D = \frac{\sum D}{n}$

Null Hypothesis $H_0$: $\mu_D = 0$

$$t = \frac{M_D - \mu_D}{s_{M_D}}$$

$$\text{variance of D} = s^2 = \frac{SS}{df}$$

$$s_{M_D} = \sqrt{\frac{s^2}{n}}$$

$$estimated\ Cohen's\ d = \frac{M_D}{s}$$

$$r^2 = \frac{t^2}{t^2 + df}$$

Confidence Interval (C.I.):

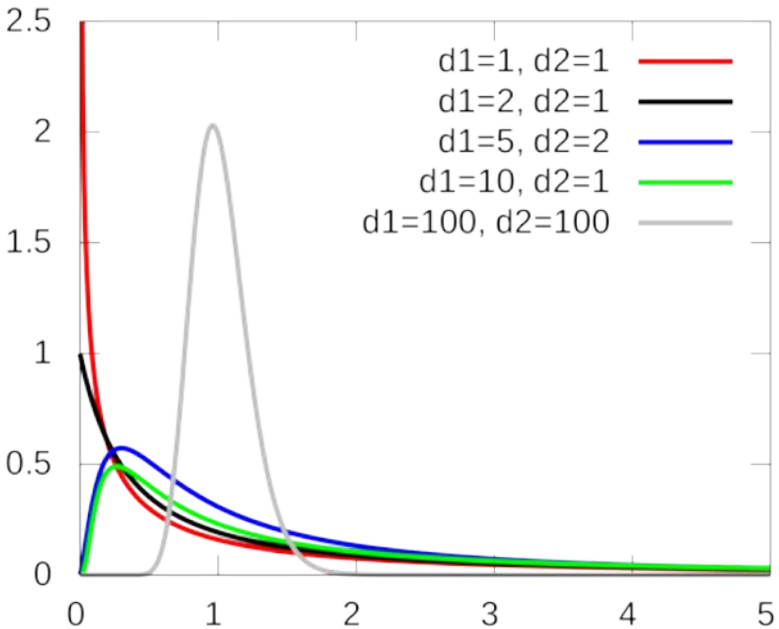
$$C.I.(1-\alpha): \mu_D = M_D \pm t_\alpha s_{M_D}$$

# F-Distribution (Fisher–Snedecor)

F-distribution
- right-skewed distribution used most commonly in Analysis of Variance.
- ( https://en.wikipedia.org/wiki/F-distribution )
- named after Ronald Fisher and George W. Snedecor

Ronald Fisher 1890-1962
British. Developed the
analysis of variance ANOVA

George W. Snedecor
1881-1974, USA




A random variate of the F-distribution with parameters d1, d2 arises as the ratio of two appropriately scaled chi-squared variates:

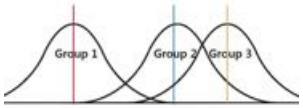
$$X = (U_1/d_1) / (U_2/d_2)$$

where $U_1$ and $U_2$ are independent and have chi-squared distributions with $d_1$ & $d_2$ degrees of freedom respectively.

## F-tests:

where the null hypothesis is that two independent normal variances are equal,
and the observed sums of some appropriately selected squares are then examined to see whether their ratio is significantly incompatible with this null hypothesis.

# ANOVA = Analysis of Variance



Ronald Fisher 1890-1962
British. Developed the analysis of variance **ANOVA**

**ANOVA** - a collection of statistical models and estimation procedures (such as the "variation" among and between groups)
used to analyze the differences among group means in a sample.

The observed variance in a particular variable is partitioned into components attributable to different sources of variation.

In its simplest form, **ANOVA** provides a statistical test of whether two or more population means are equal, and therefore generalizes the t-test beyond two means.

Variance Between
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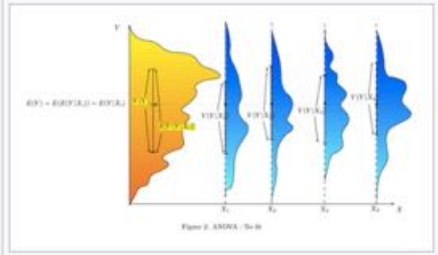
Variance Within

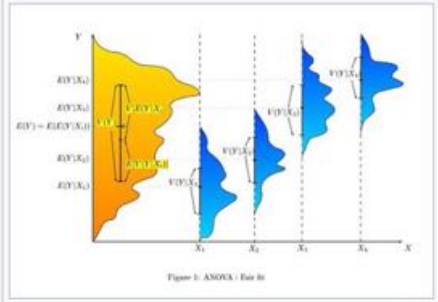
**ANOVA** is the synthesis of several ideas and it is used for multiple purposes.
As a consequence, it is difficult to define concisely or precisely.

"Classical" **ANOVA** for balanced data does three things at once:
1. Uses additive data decomposition for exploratory data analysis. Uses sums of squares to indicate the variance of each component of the decomposition.
2. Comparisons of mean squares, along with an F-test ... allow testing of a nested sequence of models.
3. Closely related to the **ANOVA** is a linear model fit with coefficient estimates and standard errors.

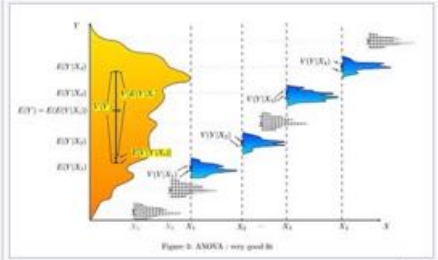
Additionally, **ANOVA** :
1. is computationally elegant and relatively robust against violations of its assumptions.
2. provides strong (multiple sample comparison) statistical analysis.
3. has been adapted to the analysis of a variety of experimental designs.



No fit: Young vs old, and short-haired vs long-haired



Fair fit: Pet vs Working breed and less athletic vs more athletic



Very good fit: Weight by breed

## More links:

Confidence Interval
- https://en.wikipedia.org/wiki/Confidence_interval

Cluster Analysis (finding groups (clusters) in data)
- https://en.wikipedia.org/wiki/Cluster_analysis

Student's t-distribution - estimating the mean for small sample sizes when **σ** is unknown.
- https://en.wikipedia.org/wiki/Student%27s_t-distribution

Chi-squared distribution - distribution of a sum of the squares of k independent standard normal random variables
- https://en.wikipedia.org/wiki/Chi-squared_distribution

F-distribution (also ANOVA - Analysis of Variance)
- https://en.wikipedia.org/wiki/F-distribution

Gamma distribution - a two-parameter family of continuous probability distributions.
Special cases – exponential and chi-squared distributions.
- https://en.wikipedia.org/wiki/Gamma_distribution

Convolution, Functional Analysis
– https://en.wikipedia.org/wiki/Convolution –
– https://en.wikipedia.org/wiki/Functional_analysis –

Extracting signal from Noise
– https://en.wikipedia.org/wiki/Signal-to-noise_ratio –
– https://en.wikipedia.org/wiki/Noise_reduction – (filtering, etc.). If we add N samples "s" of noisy data, then the signal will add proportionally to N, whereas noise will grow only as sqrt(N). So, if we add 100 samples, we will improve out Signal/Noise ratio ~10 times.

Monte Carlo Method - used to evaluate something by using randomly-generated samples.
The more samples – the more accurate the result.
- https://en.wikipedia.org/wiki/Monte_Carlo_method –

## More links:

Stochastic Processes, Time Series Analysis
- https://en.wikipedia.org/wiki/Stochastic_process –
- https://en.wikipedia.org/wiki/Time_series –
- https://en.wikipedia.org/wiki/Random_walk –
- https://en.wikipedia.org/wiki/Brownian_motion –
- https://en.wikipedia.org/wiki/Diffusion ,
- https://en.wikipedia.org/wiki/Diffusion_process –
- https://en.wikipedia.org/wiki/Stationary_process –
- https://en.wikipedia.org/wiki/Stable_process –

Poisson Process
– https://en.wikipedia.org/wiki/Poisson_point_process –
– https://en.wikipedia.org/wiki/Dirac_delta_function –
– https://en.wikipedia.org/wiki/Compound_Poisson_process –

White Noise, Gaussian Noise, Brownian Noise
– https://en.wikipedia.org/wiki/White_noise –
– https://en.wikipedia.org/wiki/Gaussian_noise –
– https://en.wikipedia.org/wiki/Brownian_noise –

Markov Process
– https://en.wikipedia.org/wiki/Markov_chain –
– https://en.wikipedia.org/wiki/Markov_process –

Correlation function:
– https://en.wikipedia.org/wiki/Correlation_function –
– https://en.wikipedia.org/wiki/Autocorrelation –

Spectral Density, Fourier Analysis
– https://en.wikipedia.org/wiki/Spectral_density –
– https://en.wikipedia.org/wiki/Fourier_analysis –
– https://en.wikipedia.org/wiki/Fast_Fourier_transform –